Multiobjective Simulated Annealing-Based Stopwords Substitution for Rubbish Text Attack

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***Abstract*—Modern natural language processing (NLP) models exhibit extreme sensitivity toward text adversarial examples, while their opposite insensitivity to text rubbish examples is greatly underestimated. Text rubbish examples usually refer to highly modified sentences that appear nonsensical to humans but can keep the model's prediction unchanged, which are significant in model robustness evaluation, improvement, and interpretation. Existing methods usually design a single objective optimization method to simultaneously maximize the modification rate and the model confidence (MC) with some text modification strategies, such as word deletion and preposition substitution. However, the single objective optimization easily falls into local optima due to the conflicting objectives, and the simple text modification strategies greatly limit the diversity of rubbish examples. To address these problems, we propose a multiobjective simulated annealing-based stopword substitution (MOSA-S2) algorithm with three major merits. First, the MOSA-S2 replaces the input words with meaningless stopwords and employs importance-based composite perturbation to simulate word substitution, enhancing the quality and diversity of the rubbish sample generation. Second, we formulate a multiobjective simulated annealing method to adaptively determine the priority of word replacements, which can escape local optima with a controlled probability and balance multiple objectives via Pareto dominance. Third, we design a grammatically constrained variant to enhance the readability of rubbish text, while maximizing its semantic deviation from the original to mislead human judgment. We evaluate the effectiveness and efficiency of our method on six text datasets by attacking seven popular neural models. Extensive experimental results demonstrate the superiority of our MOSA-S2 and reveal the fact that modern NLP models may not fully comprehend the textual semantics, as they make the same prediction with even higher confidence for nonsensical text sequences.**

***Index Terms*—Adversarial machine learning, deep neural networks (DNNs), natural language processing (NLP), rubbish examples.**

I. Introduction

DEEP neural networks (DNNs) exhibit striking brittleness toward out-of-distribution inputs in the natural language processing (NLP) field. Commonly studied adversarial examples introduce small semantically invariant perturbations to an input that cause a different model prediction, which reflects the excessive sensitivity of DNNs to tiny task-irrelevant changes. This phenomenon has attracted much attention in the NLP communities and spawned a series of text adversarial attack and defense methods [1], [2], [3], [4], [5].

In contrast, text rubbish examples refer to heavily modified examples, which are nonsensical to humans but have no effect on the model's prediction. The rubbish examples expose another type of pathological behavior: modern DNNs are too insensitive to large task-relevant modifications of inputs. Fig. 1 lists several rubbish examples crafted for text classification tasks, where the models show the same prediction with higher confidence for meaningless sentences. The rubbish examples allow the classifier to make spurious but statistically valid decisions by fitting spurious cues in the data rather than fully understanding contextual semantics. This pathology may result from overly robust predictive features in the standard data or strong class evidence in input subsets that can be identified by pretrained models.

Motivation: What is the significance of producing high-quality text rubbish examples? First, our subsequent experiments confirm that such insensitivity generally occurs in various tasks and model architectures. Prior work [6] also highlights the vulnerability of perturbation-based interpretability methods under rubbish sample attacks. Therefore, generating higher quality rubbish samples can help better assess the robustness of models and their interpretability methods. Second, incorporating rubbish samples into the training set for model retraining can enhance a model's robustness and generalization ability. Third, the rubbish sample generation can serve as a post-hoc explanation method, providing empirical insights into the behavior of deep neural networks that lack strict theoretical guarantees. Much research in the computer vision community [7], [8], [9], [10] has raised concerns that the well-trained models suffer from undersensitivity, while there are still few related studies in the NLP domain, especially for those security-sensitive tasks, such as financial sentiment

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AG News	
Before	Vandals damage bird reserve. A disturbance free zone for nesting birds is put at risk by vandals who cut down the boundary fence.
After	Yourself until most other. A any being after for as didn't is now at few by themselves who against down the as but.
Prediction	Sci/tech (95%→99%)
SNLI	
Premise	A boy is jumping on skateboard in the middle of a red bridge.
Before	The boy skates down the sidewalk.
After	The herself herself down the can.
Prediction	Contradiction (87%→100%)

Fig. 1. Rubbish examples crafted by MOSA-S2. Well-trained models retain original prediction with even higher confidence on nonsensical text sequences.

analysis [11], spam detection [12], and toxic comment detection [13]. Therefore, exploring potential text rubbish examples is crucial to characterize undersensitivity and mitigate its potential security risks.

To explore the limitations of model interpretation methods, Feng et al. [14] formulate the input reduction (IR), which iteratively removes uninformative words from the input under invariant prediction. Specifically, IR employs beam search to optimize the word deletion procedure, aiming to simultaneously maximize the modification rate and model confidence (MC). However, the produced rubbish examples usually cause a drop in MC, whereas effective rubbish samples are not expected to affect the model's prediction. Besides, since there are no constraints on the semantics of words, the remaining words still contain semantically meaningful features to humans (e.g., a technically superb film → superb). Li et al. [6] propose a preposition substitution algorithm (AGPS) for text rubbish example generation to mitigate the decrease in MC. The AGPS employs an annealing genetic algorithm to optimize the word substitution procedure by weighting multiple targets into a single target. However, weighted optimization relies heavily on hard-to-determine coefficients, and annealing genetic optimization typically requires a large number of queries to finalize modifications, which usually does not guarantee to find the Pareto optimal solutions with high efficiency.

In this work, we propose a multiobjective simulated annealing-based stopword substitution (MOSA-S2) algorithm that effectively addresses the above limitations. Specifically, the MOSA-S2 employs the English stopwords corpus of NLTK as a set of substitution candidates to expand the search space in the first step. Each input word with practical meaning can be replaced with any meaningless candidate word, and our goal is to achieve greater semantic confusion without reducing MC. To provide sufficient options for simulating word replacement, we design the importance-based composite (i.e., hybrid of single and double) perturbations to generate new candidate solutions. In the second step, we propose to adaptively optimize the word replacement priority via a tailored multiobjective simulated annealing method, which strikes a better balance between multiple objectives. Notably, the MOSA-S2 provides a probability based on Pareto dominance to dynamically adjust the substitution order, which is significantly effective in guiding the search toward the Pareto optimal solutions. Our main contributions are summarized below.

1) We propose an innovative multiobjective simulated annealing-based stopword substitution algorithm for rubbish text attacks. The MOSA-S2 replaces words with nonsensical stopwords to confuse text semantics, which greatly enhances the quality and diversity of generated rubbish samples.
2) We design an effective multiobjective simulated annealing method to prioritize word replacements, which achieves simultaneous maximization of word modification rate and MC with fewer queries.
3) We further design a grammatically constrained replacement strategy to enhance the readability and grammatical coherence of rubbish text, maximizing its semantic deviation from the original to mislead human judgment without altering the model's high confidence predictions.
4) We evaluate the effectiveness of our MOSA-S2 on six public datasets by attacking seven representative models. Extensive experiments demonstrate its superiority in generation quality and efficiency, systematically validate its robustness to candidate lexicon scales, and confirm its utility in retraining and transfer attacks.
5) Extended evaluations on large language models (LLMs) reveal that undersensitivity is a pervasive vulnerability, not limited to conventional DNNs. As a unique vulnerability class, rubbish text attacks provide a crucial complement to existing LLM safety research.

II. Related Work

In this section, we briefly discuss related work on textual adversarial attacks and rubbish text attacks.

A. Text Adversarial Attack

The generation of adversarial examples in NLP has attracted considerable attention, and researchers have proposed numerous approaches using various subtle semantically invariant text transformations for different attack tasks. Papernot et al. [1] first focus on the adversarial samples for text classification models and craft adversarial input sequences for recurrent neural networks (RNNs). Ebrahimi et al. [15] adopt a token-flipping method based on the gradients of the one-hot input vectors to achieve the adversarial attack. Li et al. [16] apply the BERT-masked language model to generate substitutes with semantic preservation. Guo et al. [17] propose the first general gradient-based adversarial attack against text transformers, defining a parameterized adversarial distribution by a continuous-valued matrix and optimizing it with a gradient-based approach. Wang et al. [18] present a unified and effective semantic adversarial attack framework SemAttack, which constructs different semantic perturbation functions to generate natural adversarial text. With the increasing deployment of LLMs, various security vulnerabilities have been identified, such as adversarial attacks [19], [20] and

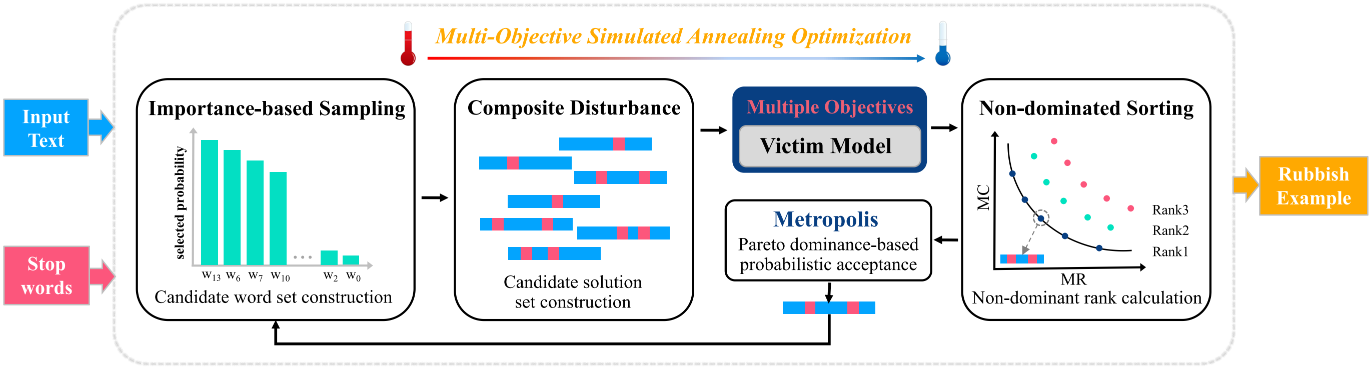

Fig. 2. Illustration of MOSA-S2. The rubbish example generation follows the multiobjective simulated annealing framework, iteratively executing the procedure at each temperature until reaching the modification limit. In the nondominated sorting module, the solution set is divided into multiple Pareto nondominated fronts, with each solution assigned a nondomination rank. The metropolis module offers an acceptance probability based on the nondomination rank to facilitate escaping local optima.

jailbreak attacks [21], [22]. These attacks use crafted prompts to bypass safety alignments and induce unintended behaviors. In this work, we investigate whether LLMs suffer from the same undersensitivity issue as conventional DNNs do, thereby complementing existing research on jailbreaking and textual adversarial attacks.

B. Rubbish Examples

As a failure mode that complements the oversensitivity of adversarial samples, the undersensitivity of rubbish samples is greatly underestimated. The concept of rubbish samples is primarily proposed in the field of computer vision by fooling images [23], [24]. Humans are completely unable to recognize these images, while the DNNs almost certainly classify these degenerate inputs as a certain category. They demonstrate that both shallow linear models and deep networks have the problem of yielding more extreme predictions at points far away from the training data. Carter et al. [9] observe that classifiers trained on CIFAR-10 and ImageNet make confident predictions on the remaining pixel subsets that even mask 95% of input images. They define this pathology as overinterpretation and attribute it to spurious statistical signals in the benchmark data. In exploring the limitations of the NLP interpretation methods, Feng et al. [14] generate text rubbish samples by iteratively removing the least important words from the input under the guarantee of constant prediction. As word deletions proceed, valid inputs are gradually transformed into rubbish samples containing only a few words, which lack human-understandable semantic features but keep the model's original prediction. They approximate word importance with input gradients and augment input reduction via beam search. However, the resulting rubbish examples usually cause a decrease in the model's confidence and contain semantically salient class-evidence. To address this issue, previous research [6] has employed a preposition-based substitution strategy to craft rubbish samples, which collects 37 commonly used prepositions as candidate substitutions and replaces input words with these prepositions. However, the annealing genetic algorithm used to determine word replacement priorities still employs single-objective weighted optimization to handle multiobjective optimization. This approach is complex and subjective in weight selection and struggles to obtain a widely distributed Pareto optimal set. Additionally, the annealing genetic algorithm searches the rubbish examples in the optimization space by performing heuristic operations, which requires an abundance of queries to identify the final modifications, resulting in high computational costs and low efficiency.

III. Methodology

This section details the proposed MOSA-S2 algorithm, with its workflow illustrated in Fig. 2. Before delving into details, we discuss the definitions of rubbish text attack and multiobjective optimization.

A. Problem Definition

Given an input space containing N samples $\mathcal{X} = \{\mathbf{X}_i\}_{i=1}^{N}$ and an output space $\mathcal{Y} = \{\mathbf{Y}_i\}_{i=1}^{L}$ containing L labels, the classifier F needs to learn a mapping $f\colon \mathcal{X} \rightarrow \mathcal{Y}$ so that the input sample $\mathbf{X}$ can be classified to the correct label $\mathbf{Y}_{true}$, by maximizing the posterior probability

$$\underset{\mathbf{Y}_i \in \mathcal{Y}}{\operatorname{argmax}}\, P\left(\mathbf{Y}_i | \mathbf{X}\right) = \mathbf{Y}_{true}. \tag{1}$$

A reasonable text rubbish example attacker pursues to make a heavy perturbation on the input text $\mathbf{X} = \{w_1, w_2, \ldots, w_n\}$, where the rubbish example $\mathbf{X}^*$ is unreadable to humans but maintains the classifier's prediction

$$\underset{\mathbf{Y}_i \in \mathcal{Y}}{\operatorname{argmax}}\, P\left(\mathbf{Y}_i | \mathbf{X}^*\right) = \mathbf{Y}_{true}. \tag{2}$$

The rubbish text attack task can be transformed into a multiobjective constrained optimization problem. Under the constraint of constant model prediction, the objectives are maximizing the difference between input $\mathbf{X}$ and rubbish output $\mathbf{X}^*$ and simultaneously ensuring the classifier preserves high confidence in the original label

$$\mathcal{J}_1\left(\mathbf{X}, \mathbf{X}^*\right) = \texttt{len}\left(\mathbf{X}^* \neq \mathbf{X}\right) \tag{3a}$$

$$\mathcal{J}_2\left(F, \mathbf{X}^*\right) = F\left(\mathbf{X}^*\right) \tag{3b}$$

$$\text{s.t. } \mathbf{Y} = \mathbf{Y}_{true}. \tag{3c}$$

Some basic definitions of the multiobjective optimization problem in this work are as follows.

1) *Dominance:* $\mathbf{X}_p$ is said to dominate $\mathbf{X}_q$, which is denoted by $\mathbf{X}_p \prec \mathbf{X}_q$, if $\mathcal{J}_i(\mathbf{X}_q) \leq \mathcal{J}_i(\mathbf{X}_p)$ for all $i = 1, 2$, and $\mathcal{J}_j(\mathbf{X}_q) < \mathcal{J}_j(\mathbf{X}_p)$ for at least one in $j = 1, 2$.
2) *Pareto Optimality:* If no solution in the solution set can dominate $\mathbf{X}^*$, it is called Pareto optimal.
3) *Pareto Set (PS):* The Pareto optimal set is defined as the set of all Pareto optimal solutions in the decision space.
4) *Pareto Front (PF):* The set of all objective vectors in PS is called the Pareto optimal front.

To achieve this multiobjective optimization, we need to address two problems: 1) how to select substitution candidates and 2) how to determine the word attack priority.

B. Stopwords-Based Word Substitution

Stopwords are common high-frequency words that contain no practical semantic information. In NLP tasks, stopwords are generally removed from the analyzed text to improve the quality of text features or narrow the dimension of word-based features. To produce human-unreadable modifications, we propose to employ stopwords to replace task-related words. To this end, we use the English stopwords corpus of NLTK as a set of meaningless substitutions $\mathbb{S}$. The part-of-speech (POS) tagging for each word needs to be determined before the substitution procedure. To achieve efficiency, only four kinds of input words (i.e., nouns, adjectives, verbs, and adverbs) with practical meaning can be replaced, and all the other words are filtered out.

For each modifiable word w_i, every substitution word $w_i' \in \mathbb{S}$ is a potential candidate. To search for the best candidate, the candidate importance score $S_{w_i'}$ for each w_i' is defined as the original label prediction probability

$$S_{w_i'} = P\left(\mathbf{Y}_{true} | \mathbf{X}_i'\right) \quad \forall w_i' \in \mathbb{S} \tag{4}$$

where

$$\mathbf{X}_i' = \left\{w_1, w_2, \ldots, w_i', \ldots, w_n\right\}. \tag{5}$$

Then, we pick the candidate w_i' that maximizes the probability of the original label as the best substitution word w_i^*, which is formulated in the following equation:

$$w_i^* = \underset{w_i' \in \mathbb{S}}{\operatorname{argmax}}\, S_{w_i'}. \tag{6}$$

We implement this process for the word substitution to solve the first issue.

C. Multiobjective Simulated Annealing for Priority Determination

1) Gradient-Based Word Importance Ranking: The word importance score of w_i reflects the contribution of w_i to the model's prediction. MOSA-S2 employs a gradient-based method to calculate the importance of w_i by the gradient of the loss function, as shown in the following equation:

$$I(w_i) = \|\nabla_{\mathbf{e_i}} \mathbf{L}(\theta, \mathbf{X}, \mathbf{Y}_{true})\|_1 \tag{7}$$

where $\mathbf{L}(\theta, \mathbf{X}, \mathbf{Y}_{true})$ is the loss function for input text $\mathbf{X}$ with label $\mathbf{Y}_{true}$, and e_i represents the word embedding corresponding to word w_i. For models where inputs are tokenized into subwords, we average the gradients of all tokens constituting the word to calculate the importance of each word. The gradient-based method requires only one forward–backward pass in calculation and outperforms other methods (e.g., deletion-based and PWWS-based) in balancing the attack performance and efficiency [25].

Adversarial attacks focus on important words, while rubbish example generation prioritizes unimportant words. A straightforward way is to substitute words in the ascending order of word importance scores. Formally, the following equation gives the reordered sequence:

$$O = \{I_1 < I_2 < \ldots < I_n\}. \tag{8}$$

However, sequentially replacing words with this static order easily leads to local optima and triggers the early flipping of model prediction, as there is no theoretical guarantee that the top-k unimportant words are the best combination for keeping the classifier's prediction. For example, modifying three words $\{top1, top3, top2\}$ in O may perform better at maintaining predictions than the two-word modification $\{top1, top2\}$. Therefore, it is critical to optimize word substitution priority.

2) Multiobjective Simulated Annealing Optimization: Before detailing our algorithm, we first explain the concepts of the original simulated annealing (SA) [26]. The SA is an intelligent optimization algorithm inspired by the physical annealing procedure of metals. At a given initial temperature, as the temperature decreases, SA combines probabilistic jump characteristics to search for the global optimal solution of the objective function. SA has been widely used in the optimization field because of its unique optimization mechanism, less dependence on problem information, and strong flexibility. In a multiobjective problem, the objectives may conflict with each other and cannot be directly compared to those of a single-objective problem. Therefore, SA needs to be extended to suit multiobjective optimization problems.

In this work, we propose a tailored multiobjective simulated annealing algorithm to determine attacking priority. Specifically, MOSA-S2 has a probability related to the nondomination rank to guide the search toward the Pareto optimal solution set and escape from local optima. Besides, the importance-based composite perturbations are significant in providing sufficient options for simulating word replacements. The MOSA-S2 search starts from line 2 of Algorithm 1 with the highest temperature T_0. At each temperature, the optimization procedure has two main components: 1) generation of new candidate solutions based on word importance and 2) probabilistic acceptance based on Pareto dominance. Next, we describe our algorithm step by step.

In the first step, we first employ the gradient-based approach to calculate the word importance $I(w_i)$ in (7) and reorder them to get $\mathbf{X}_r = \{w_{r0}, w_{r1}, \ldots, w_{r(n-1)}\}$ in line 3. Then, we repeatedly sample K times from the words in $\mathbf{X}_r$ to generate a replacement candidate set $\mathbb{S}_t$ as the new search space in line 4. To make the unimportant words have a higher probability of being selected, the probability P_i of the word w_{ri} in $\mathbf{X}_r$

Algorithm 1 Proposed MOSA-S2 Algorithm

Input: Initial sentence with n words $\mathbf{X} = (w_1, \ldots, w_n)$

Input: Well-trained classification model F

Output: Rubbish example $\mathbf{X}^*$

1 Initialization: the initial rubbish example $\mathbf{X}^* = \mathbf{X}$, the initial non-domination rank $R_{\mathbf{X}^*} = \infty$, the initial time $t = 0$, the initial temperature $T = T_0 = 1000$, the attenuation factor $\alpha = 0.9$, the lowest temperature $T_{min} = \alpha^n \cdot T_0$, the internal simulation steps $K = 15$, and the sampling radius $\delta = 2$;

/* The MOSA-S2 search starts */

2 **while** $T > T_{min}$ **do**

/* Create candidate solutions */

3 $\mathbf{X}_r \leftarrow$ Rank words in $\mathbf{X}^*$ in ascending order of importance;

4 Craft candidate word set $\mathbb{S}_t$ for substitution;

5 $\mathbb{C}^t = \mathcal{P}(\mathbf{X}^*, \mathbf{X}, F, \mathbb{S}_t)$; // $\mathcal{P}$ is defined in Algorithm 2

/* Probabilistic acceptance */

6 $\mathbb{F}^t \leftarrow$ Rank the members of $\mathbb{C}^t$ via the fast non-dominated sorting;

7 $\mathbb{F}^t \leftarrow$ Randomly shuffle $\mathbb{F}^t$;

8 **for** $\mathbf{X}_{new}$ *in* $\mathbb{F}^t$ **do**

9 **if** $R_{\mathbf{X}_{new}} < R_{\mathbf{X}^*}$ **then**

10 $\mathbf{X}^* = \mathbf{X}_{new}$;

11 **else**

12 $p = e^{-((R_{\mathbf{X}_{new}} - R_{\mathbf{X}^*}) \cdot \beta)/T}$;

13 $r = random(0, 1)$;

14 **if** $r < p$ **then**

15 $\mathbf{X}^* = \mathbf{X}_{new}$;

16 $t = t + 1$;

17 $T = \alpha \cdot T$; // Annealing

18 **return** Rubbish example $\mathbf{X}^*$

being chosen is inversely proportional to its importance in the following equation:

$$P_i = \begin{cases} \dfrac{1 - \frac{i}{n}}{\sum_{i=0}^{n-1} \left(1 - \frac{i}{n}\right)}, & \text{if } i \leq \delta \cdot t \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where δ is the sampling radius, which allows us to modify the least important word at the initial time ($t = 0$) and enlarges the sampling radius over time. Then, we design the word importance-based perturbation function $\mathcal{P}$ to construct the new candidate solution set $\mathbb{C}^t$ in line 5, as shown in Algorithm 2. Particularly, this procedure combines single and double perturbations to explore the neighborhood of the current solution. We visit each *Candidate* of its set $\mathbb{S}_t$, and replace it in the current sentence $\mathbf{X}^*$ to create the perturbed example $\mathbf{X}_P$. Then, we query the victim model F to obtain the prediction of $\mathbf{X}_P$. If its predicted label is the same as the original label, we append $\mathbf{X}_P$ to the new candidate solution set $\mathbb{C}^t$. To achieve efficiency, the current *Candidate* is removed from $\mathbb{S}_t$ before the second perturbation. Similarly, we iteratively modify *Candidate* in

Algorithm 2 Perturbation Function $\mathcal{P}$

Input: Current sentence $\mathbf{X}^*$, Initial sentence $\mathbf{X}$

Input: Well-trained classification model F

Input: Candidate substitution set $\mathbb{S}_t$

Output: New candidate solution combinations $\mathbb{C}^t$

Initialize $\mathbb{C}^t = \varnothing$;

for $Candidate$ *in* $\mathbb{S}_t$ **do**

$\mathbf{X}_P \leftarrow$ Craft a perturbed example by replacing $Candidate$ in $\mathbf{X}^*$;

if $F(\mathbf{X}_P) = F(\mathbf{X})$ **then**

$\mathbb{C}^t = \mathbb{C}^t \cup \{\mathbf{X}_P\}$;

$\mathbb{S}_{t2} \leftarrow \mathbb{S}_t \setminus Candidate$;

for $Candidate$ *in* $\mathbb{S}_{t2}$ **do**

$\mathbf{X}_{P2} \leftarrow$ Craft a perturbed example by replacing $Candidate$ in $\mathbf{X}_P$;

if $F(\mathbf{X}_{P2}) = F(\mathbf{X})$ **then**

$\mathbb{C}^t = \mathbb{C}^t \cup \{\mathbf{X}_{P2}\}$;

return $\mathbb{C}^t$

$\mathbf{X}_P$ by traversing the updated candidate set $\mathbb{S}_{t2}$ to yield the secondary perturbed sample $\mathbf{X}_{P2}$, and add the sample with unaltered prediction to $\mathbb{C}^t$.

The second step is to determine whether the modified example is accepted or not at the current temperature in lines 6–15. Specifically, we follow the fast nondominated sorting operator in NSGA-II [27] to rank the members of the current candidate solutions set $\mathbb{C}^t$ in line 6. By sorting, the solutions set is divided into multiple Pareto nondominated sets $\mathbb{F}^t$, and each solution $\mathbf{X}^*$ is assigned a nondomination rank $R_{\mathbf{X}^*}$. The classical SA uses the Metropolis probability to accept or refuse one solution. We propose to combine the Metropolis principle with the Pareto nondominated sorting, where the acceptance probability depends on the nondomination rank of each solution. The probability p is defined by

$$p = e^{-((R_{\mathbf{X}_{new}} - R_{\mathbf{X}^*}) \cdot \beta)/T} \tag{10}$$

where β is a constant to adjust the value to the appropriate range. If the nondomination rank of the new solution $\mathbf{X}_{new}$ is lower than that of the current solution $\mathbf{X}^*$ (i.e., $\mathbf{X}_{new} \prec \mathbf{X}^*$), we directly accept it as a good perturbed example. Otherwise, we probabilistically accept a worse solution by (10) as listed in lines 12–15. After the internal iteration, time t increases (line 16), and the temperature T reduces (line 17). We empirically adopt the proportional cooling function

$$T = \alpha \cdot T \tag{11}$$

where the attenuation factor $\alpha = 0.9$. This process continues until the temperature drops to T_{min}. To save the computation cost, we adaptively adjust T_{min} according to the length of the initial input (i.e., n)

$$T_{min} = \alpha^n \cdot T_0. \tag{12}$$

This solves the second key problem.

IV. Experiments

We provide the source code on GitHub[1] to ensure that all the results in this section are reproducible.

A. Datasets and Victim Models

To verify the attack performance of our MOSA-S2 in different tasks, we investigate six text classification datasets with various properties, including MR [28] and IMDB [29] for sentiment analysis, AG News [30] for news topic classification, SNLI [31] for natural language inference, and MRPC [32], QQP [33] for paraphrasing. The dataset statistics are summarized in Table I.

1) *MR:* The binary classification dataset to evaluate whether movie reviews express positive or negative sentiment.
2) *IMDB:* The large binary sentiment analysis dataset with positive and negative classes.
3) *AG News:* The collection of news articles covering four categories: world, sports, business, and sci/tech.
4) *SNLI:* The tripartite classification dataset to predict the relationship between premise and hypothesis, that is, entailment, neutral, and contradiction.
5) *MRPC:* The semantic similarity classification dataset to deduce the semantics of the context sentence as equivalent or not equivalent.
6) *QQP:* The question-pair paraphrase identification dataset with binary classes, that is, duplicate and not duplicate.

We evaluate our method by attacking seven models with different architectures and sizes, namely CNN, LSTM, BERT (bert-base-uncased) [34], RoBERTa (roberta-base) [35], ALBERT (albert-base-v2) [36], DistilBERT (distilbert-base-cased, distilbert-base-uncased) [37], and XLNet (xlnet-base-cased) [38]. The DistilBERT-uncased is case insensitive, and all tokens are in lowercase. Different versions of DistilBERT are implemented on different datasets, that is, (cased: SNLI, QQP) and (uncased: MRPC). All these models can be downloaded from HuggingFace.[2]

B. Baseline Methods

To evaluate the effectiveness of our method, we compare MOSA-S2 with existing text rubbish example generation methods, such as **Input Reduction** [14] and **AGPS** [6]. As discussed in Section II, previous methods encounter several significant challenges. First, prior studies implemented single-objective weighted optimization to handle two interacting objectives, making it difficult to find Pareto optimal solutions. Moreover, existing techniques for word modification usually lead to a decrease in MC and retaining semantically salient content.

C. Evaluation Metrics and Experimental Setup

1) Evaluation Metrics: We evaluate the attack performance considering four metrics, that is, the MC, the modification rate (MR), the perplexity (PPL), and the semantic similarity (SIM). The MC is defined as the true label probability of the classifier for the rubbish sample. MR measures the proportion of modified words among all eligible words in the sentence. Considering that only semantic input words are replaced (i.e., nouns, adjectives, verbs, and adverbs), only these words are taken into consideration when reporting MR. The PPL serves to evaluate the fluency of texts [2], [39], computed using a small-sized GPT-2 model with a 50K vocabulary [40]. The SIM refers to the semantic similarity between the original input and the rubbish sample. Following [41] and [42], the universal sentence encoder (USE) [43] with cosine similarity is applied to calculate the semantic similarity before and after attacks. Intuitively, a rational attacker aims to achieve a high MR, a low SIM, and a high PPL with a high MC.

2) Experimental Setup: We list the fine-tuned parameter settings for our MOSA-S2 in line 1 of Algorithm 1. For the baselines, we adhere to their protocols and adopt the author-recommended parameter values to ensure a fair comparison. To achieve efficiency, we randomly select 500 instances from each dataset to implement the text attack. Especially in the natural language inference task (i.e., SNLI), we only modify the hypothesis, holding the premise unchanged. All experiments are conducted on the TextAttack framework [44].

TABLE I

Statistic Information of the Six Datasets. "# Words" Denotes the Average Number of Words

Dataset	# Train	# Test	# Classes	# Words
MR	8530	1066	2	18.49
IMDB	25000	25000	2	235.72
AG News	120000	7600	4	38.78
SNLI	550152	10000	3	20.27
MRPC	3668	1725	2	39.02
QQP	363846	390965	2	22.24

D. Experimental Results

The experimental results of average MC, MR, PPL, and SIM are shown in Tables II and III. We demonstrate our contributions as claimed in the Introduction by asking the following three questions.

Q1: Is the stopwords-based substitution superior to other modification methods in keeping the model's confidence? To alleviate the decline of MC, we adopt the word replacement attack strategy and utilize the stopwords corpus to enrich our substitution candidates. The results of MC in Table II indicate that our MOSA-S2 consistently outperforms the two baselines by a significant margin on all victim models and datasets, achieving an average MC improvement of 11.39% over IR and 3% over AGPS, with the advantage being particularly evident on CNN. Besides, the rubbish samples crafted by MOSA-S2 achieve a higher average MC (by 4.53%) than the original input on all victim models. This strongly validates the benefit of stopwords-based substitution.

[1] https://github.com/soar-create/MOSA-S2

[2] https://huggingface.co/models

TABLE II

AVERAGE MC, THE AVERAGE MR, AND THE PERPLEXITY (PPL) OF DIFFERENT ALGORITHMS ON SIX TEXT CLASSIFICATION DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. THE "ACC" COLUMN SHOWS THE ORIGINAL ACCURACY OF MODELS, AND THE "ORIGINAL" COLUMN DENOTES THE ORIGINAL AVERAGE MC WITHOUT ATTACKS

Dataset	Model	ACC	Length	MC ↑				MR ↑			PPL ↑			
				Original	IR	AGPS	MOSA-S2	IR	AGPS	MOSA-S2	Original	IR	AGPS	MOSA-S2
MR	CNN	98.00%	18.62	88.54%	67.29%	93.99%	**98.27%**	92.34%	94.84%	**95.17%**	72.70	75.53	122.78	**143.25**
	LSTM	89.60%	18.62	86.36%	91.08%	95.30%	**99.10%**	92.58%	94.66%	**95.42%**	72.70	65.02	122.53	**148.21**
	BERT	99.80%	18.62	99.68%	98.28%	98.56%	**99.94%**	92.36%	93.70%	**95.15%**	72.70	82.07	120.89	**145.60**
	RoBERTa	95.20%	18.62	94.38%	90.89%	98.37%	**99.25%**	91.28%	94.64%	**95.21%**	72.70	81.40	122.69	**145.93**
	AlBERT	93.20%	18.62	93.18%	91.40%	87.84%	**97.45%**	92.45%	95.09%	**95.32%**	72.70	83.56	124.63	**147.82**
	XLNet	98.20%	18.62	97.68%	96.79%	99.24%	**99.83%**	92.33%	94.76%	**95.25%**	72.70	82.31	122.72	**146.98**
IMDB	CNN	83.75%	227.78	90.65%	97.95%	94.44%	**100.00%**	89.20%	71.40%	**91.79%**	37.19	38.36	115.30	**118.62**
	BERT	99.80%	227.78	99.80%	95.14%	99.22%	**99.96%**	88.95%	60.32%	**91.98%**	37.19	37.18	101.69	**109.18**
AG News	CNN	88.00%	41.76	89.08%	56.68%	85.48%	**95.37%**	94.45%	96.18%	**97.34%**	62.28	167.93	176.66	**188.61**
	LSTM	86.20%	41.76	89.99%	80.93%	91.65%	**97.00%**	94.52%	95.86%	**97.28%**	62.28	119.67	157.36	**186.92**
	BERT	99.80%	41.76	99.23%	98.64%	99.07%	**99.55%**	96.95%	96.24%	**97.24%**	62.28	161.49	181.52	**188.89**
SNLI	BERT	95.60%	22.19	96.17%	87.07%	98.64%	**99.02%**	75.88%	98.75%	**99.36%**	4.11	3.34	5.73	**6.82**
	DistilBERT	86.80%	22.19	91.40%	83.94%	92.45%	**96.39%**	73.69%	98.39%	**99.19%**	4.11	3.64	5.69	**6.27**
	AlBERT	92.60%	22.19	92.88%	78.51%	95.73%	**96.84%**	78.35%	98.49%	**99.23%**	4.11	3.58	5.68	**6.74**
MRPC	BERT	95.40%	39.03	94.55%	82.21%	92.74%	**98.12%**	95.16%	95.61%	**95.98%**	25.86	53.38	100.36	**106.67**
	DistilBERT	91.00%	39.03	88.54%	68.34%	91.93%	**95.02%**	90.32%	95.93%	**96.05%**	25.86	49.34	102.50	**110.57**
	XLNet	95.40%	39.03	93.53%	83.87%	96.49%	**98.70%**	94.27%	95.29%	**96.05%**	25.86	52.92	99.61	**109.27**
QQP	BERT	96.20%	22.00	97.00%	97.95%	99.53%	**99.65%**	89.20%	89.29%	**96.84%**	30.16	80.09	68.60	**80.30**
	DistilBERT	95.20%	22.00	97.06%	95.14%	98.26%	**99.19%**	88.95%	90.05%	**96.72%**	30.16	81.09	71.62	**82.75**
	AlBERT	98.20%	22.00	98.18%	98.56%	99.50%	**99.71%**	88.62%	89.38%	**95.97%**	30.16	79.54	69.72	**79.70**
Average		—	—	93.89%	87.03%	95.42%	**98.42%**	89.59%	91.94%	**96.13%**	43.89	70.07	99.91	**112.96**

TABLE III

EFFICIENCY COMPARISON BETWEEN AGPS AND MOSA-S2 WITH AN EXPANDED SET OF SUBSTITUTION CANDIDATES INCLUDING STOPWORDS. "QUERY" INDICATES THE NUMBER OF TIMES TO QUERY FOR AN EXAMPLE

Dataset	ACC	Length	Queries ↓		MC ↑			MR ↑		PPL ↑			SIM ↓	
			AGPS	MOSA-S2	Original	AGPS	MOSA-S2	AGPS	MOSA-S2	Original	AGPS	MOSA-S2	AGPS	MOSA-S2
MR	95.67%	18.62	25680	**5450**	93.30%	96.72%	**98.97%**	62.63%	**95.17%**	72.70	122.71	**146.30**	**0.57**	0.58
IMDB	91.78%	227.78	77637	**25154**	95.22%	99.52%	**99.98%**	69.90%	**91.89%**	37.19	108.50	**113.90**	0.72	**0.63**
AG News	91.33%	41.76	43991	**13390**	92.77%	92.71%	**97.30%**	96.96%	**97.29%**	62.28	171.51	**188.14**	0.69	**0.68**
MRPC	93.93%	39.03	39633	**11816**	92.21%	96.80%	**97.28%**	95.95%	**96.03%**	25.86	100.82	**108.84**	**0.66**	**0.66**
QQP	96.53%	22.00	30235	**6621**	97.41%	99.25%	**99.48%**	95.42%	**96.51%**	30.16	69.98	**80.92**	**0.65**	**0.65**

Q2: Whether the multiobjective simulated annealing is closer to the Pareto optimal solutions compared to single-objective weighted optimization approaches?

The common goal of these optimizations is to achieve greater semantic confusion without reducing MC, reflected in the evaluation criteria by the simultaneous maximization of MC and word modification rate. Experimental results in Table II show that MOSA-S2 improves MR by 6.54% compared to IR and by 4.19% compared to AGPS. As can be observed from the overall results, our MOSA-S2 accomplishes both the highest MC and MR, which indicates the superiority of our method in escaping local optima and balancing multiple objectives.

Q3: Can our MOSA-S2 enhance the quality and efficiency of the rubbish sample generation? From the PPL part in Table II, our method achieves significantly higher perplexity scores than baseline methods across all datasets, with an average improvement of 29.84% over IR and 13.05% over AGPS. The elevated PPL indicates that the rubbish samples generated by MOSA-S2 exhibit greater textual confusion and lower fluency. Results in the SIM column of Table III further show that MOSA-S2 also maintains certain advantages in semantic similarity metrics. This confirms that our approach further deceives human perception by enhancing semantic confusion and reducing semantic similarity to the original input. Moreover, we observe that IR frequently contains salient class-evidence information in the final rubbish samples. As the examples shown in Table IV, in the sentiment analysis task, the IR example, that is, "fun" strongly indicates a positive sentiment, and in the paraphrasing task, the remaining two identical words are easily judged to "Duplicate". However, our MOSA-S2 solves this issue by using substitutions with part-of-speech constraints. To further demonstrate the efficiency advantage of our MOSA-S2 over AGPS under the same modification strategy, we adjust the replacement candidate word set for AGPS from original prepositions to stopwords in the comparative experiments. Table III presents the automatic evaluation results and query consumption for rubbish sample generation. We only present the average number of queries across all victim models for each dataset to ensure

TABLE IV
RUBBISH EXAMPLES CRAFTED BY IR AND MOSA-S2

Dataset: MR Model: XLNet Label: Positive (100%)
This clever caper movie has twists worthy of David Mamet and is enormous fun for thinking audiences.
IR: Positive (100%)
fun.
MOSA-S2: Positive (100%)
This an an more has those will of David Mamet and is such can for those we.
Dataset: QQP Model: BERT Label: Duplicate (91%)
Question 1: How is technology helping us? Question 2: How can technology help us?
IR: Duplicate (99%)
Question 1: us? Question 2: us?
MOSA-S2: Duplicate (100%)
Question 1: How is same that'll us? Question 2: How can it's does us?

TABLE V
HUMAN EVALUATION ON PLAUSIBILITY AND GRAMMATICALITY FOR ORIGINAL AND GENERATED RUBBISH SAMPLES. SCORES ARE AVERAGED ACROSS THREE ANNOTATORS. THE VICTIM MODEL IS BERT

Dataset	Plausibility ↓				Grammaticality ↓			
	Orig	IR	AGPS	MOSA-S2	Orig	IR	AGPS	MOSA-S2
IMDB	4.6	2.9	1.8	1.3	4.1	2.6	1.4	1.2
SNLI	4.8	3.4	1.2	1.2	4.5	3.1	1.2	1.1
Average	4.7	3.2	1.5	1.3	4.3	2.9	1.3	1.2

clarity. The results indicate that MOSA-S2 consumes fewer queries while searching for optimal replacement priorities and simultaneously generates higher quality rubbish samples.

Overall, MOSA-S2 achieves better performance than baselines on the key metrics, indicating the merit of our method.

E. Human Evaluations

We conduct a human evaluation to empirically verify that our rubbish examples align with the core definition of being *semantically meaningless to humans*, by assessing the *Plausibility* and *Grammaticality* of the generated samples. To cover diverse task types and text lengths, we sample from IMDB and SNLI. For each dataset, we randomly select 50 examples and mix them with the generated rubbish samples before presenting them to three annotators with basic NLP knowledge. We ask the annotators to score the *Plausibility* and *Grammaticality* of each sample, scored on a 1–5 scale following established protocols [45], [46]. Specifically, Plausibility evaluates whether the text conveys interpretable semantics, where a score of 1 indicates completely illogical or semantically nonsensical text and a score of 5 indicates semantically coherent text. Grammaticality assesses syntactic readability, where a score of 1 indicates severely ungrammatical and unreadable text and a score of 5 indicates fully grammatical text. Each item receives independent ratings from three annotators, and the reported score is the mean across annotators. Results in Table V show that MOSA-S2 obtains the lowest plausibility and grammaticality scores across both datasets. This confirms that MOSA-S2 generates the most nonsensical and grammatically disrupted text, effectively fulfilling the objective of rubbish example generation by maximizing human incomprehension. The clear degradation in text quality underscores the vulnerability of models that maintain high confidence in such inputs.

TABLE VI
ABLATION RESULTS ON COMPOSITE DISTURBANCE (COMPDIST), GRADIENT-BASED IMPORTANCE RANKING (GRADRANK), TEMPERATURE-BASED PROBABILISTIC ACCEPTANCE IN SA (TPA), AND NONDOMINATED SORTING (NDS)

Component	MC	MR	PPL	SIM
w/o NDS	90.95%	91.65%	101.70	0.68
w/o CompDist	94.28%	91.58%	98.65	0.68
w/o GradRank	96.08%	93.39%	102.86	0.67
w/o TPA	94.26%	92.62%	102.42	0.67
MOSA-S2	98.01%	95.87%	106.67	0.66

V. ANALYSIS

This section presents a comprehensive analysis of MOSA-S2. We begin with an ablation study to quantify component contributions, followed by evaluating model robustness through adversarial retraining and investigating transferability across architectures. We then conduct a frequency analysis of substitution words and provide an in-depth analysis of substitution strategies.

A. Ablation Study

We conduct a quantitative ablation study to analyze the components of MOSA-S2 and summarize the results in Table VI. We choose BERT as the victim model and evaluate on 1000 randomly selected MRPC instances.

1) Without Nondominated Sorting: We replace the multiobjective framework with a single-objective simulated annealing that optimizes the weighted sum of MC and MR using AGPS-recommended weights. This simplistic aggregation scheme fails to adaptively capture the complex tradeoff between the conflicting objectives, leading to unstable performance and a bias toward a single objective. As evidenced in Table VI, this results in substantial declines in both MC (i.e., a decrease of 7.06%) and MR (i.e., a decrease of 4.22%). These results affirm that nondominated sorting is indispensable for dynamically balancing multiple objectives and achieving collaborative optimization.

2) Without Composite Disturbance: We ablate the composite perturbation strategy, allowing only single-word replacements and keeping all other settings unchanged. Without this component, at each temperature, the search relies on a single replacement pattern, which limits the modification magnitude and reduces candidate diversity. Consequently, as shown in Table VI, the attack's effectiveness is impaired,

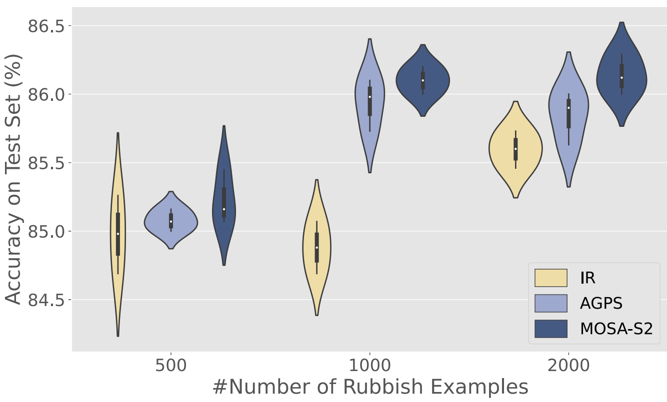

Fig. 3. Adversarial retraining results on MR dataset.

TABLE VII
DEFENSE PERFORMANCE TEST AFTER MODEL RETRAINING

Metric	Before	Retrain_{500}	Retrain_{1000}	Retrain_{2000}
MC	99.94%	98.56%	93.17%	97.14%
MR	95.15%	81.82%	81.33%	86.13%
SIM	0.58	0.63	0.61	0.61

with observable decreases in both MC and MR. This outcome indicates that the progressive composite disturbance efficiently explores the local neighborhood and yields richer candidates for subsequent nondominated ranking.

3) Without Gradient-Based Importance Ranking: We substitute the gradient-based importance ranking with a random token ordering before constructing the candidate set. Without this gradient guidance, the attack loses the awareness to prioritize less impactful words, often leading to premature modifications of critical tokens that can trigger early prediction flips and hinder the overall attack progression. The performance degradation in Table VI highlights the importance of gradient-based ranking, which orchestrates efficient attacks by preserving prediction-stable words.

4) Without Temperature-Based Probabilistic Acceptance in SA: In this variant, we remove the temperature schedule and probabilistic acceptance mechanism while retaining all other components, including the candidate word sampling and composite perturbation strategies. The annealing process is replaced with a deterministic Pareto search that only accepts candidates strictly dominating the current solution. This deterministic policy eliminates the possibility of accepting worse but potentially beneficial moves, thereby losing the ability to probabilistically escape local optima. As shown in Table VI, average MC drops by 3.75% and average MR decreases by 3.25%.

B. Model Retraining

Adversarial training is an effective technique to improve the model's robustness by joining adversarial examples into the training set. In this part, we randomly generate and add {500, 1000, 2000} MR rubbish samples to its training set and retrain the BERT model, where the rubbish examples are assigned negative weights in the adversarial loss. Then, we reattack the retrained model to evaluate the defense performance. Results in Table VII show that adversarial training mitigates this undersensitivity, as evidenced by lower modification rates, decreased MC, and the retention of more semantics in the generated rubbish examples during the reattack. Additionally, Fig. 3 shows the multiretrain classification accuracy on the test set, indicating our MOSA-S2 brings greater robustness improvement than the baselines.

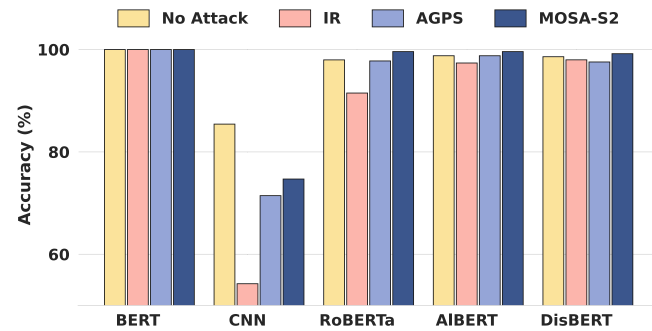

Fig. 4. Transferability test results on SST-2 dataset.

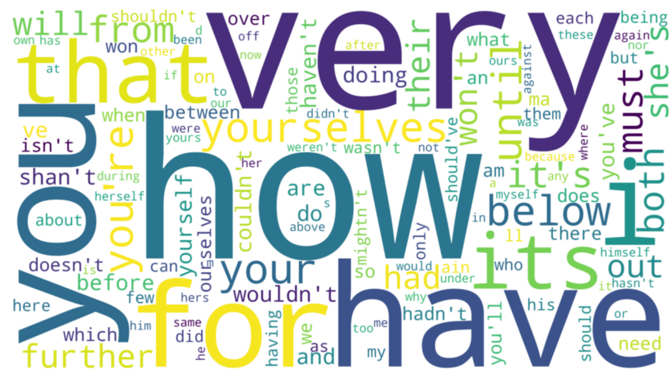

Fig. 5. Word frequency map of stopwords as replacement words appearing in generated rubbish samples.

C. Transferability

For rubbish samples, the transferability implies whether the rubbish samples curated for a certain classifier F_1 can also keep the unaltered prediction on another unknown classifier F_2. We evaluate the transferability on the binary classification dataset SST-2 [47]. Specifically, we collect rubbish examples crafted for BERT and then test the transferability on four unknown models (CNN, RoBERTa, AlBERT, and DistilBERT). Fig. 4 shows their classification accuracy on both clean data (no attack) and rubbish samples. In particular, their accuracy on BERT is 100% because the attack is prediction-based. The results manifest that the three methods exhibit strong transferability on transformers, and our MOSA-S2 is superior to the baselines in all cases.

D. Frequency Analysis of Substitution Words

We carried out a frequency analysis of replacement words to examine the occurrence of each stopword among all generated rubbish samples, as depicted in Fig. 5. From this analysis, we can see that the most frequent replacement words include "how," "you," "very," and others, while words such as "or" and "after" occur less frequently. There is significant variation in the classifier's sensitivity to different stopwords. High-frequency words signal a tendency in our algorithm to select them as replacements, as substituting input terms with these stopwords more effectively maintains the classifier's original decisions. Consequently, when launching attacks on models, high-frequency stopwords may serve as effective replacement candidates for improving attack efficiency.

E. Analysis of Substitution Strategies

This section covers the following two components: 1) the sensitivity to the size of the candidate replacement lexicon and 2) the performance of substitution strategies under grammatical and semantic constraints. Under various substitution strategies, we consistently employ BERT as the victim model and randomly sample 1000 instances from each dataset to execute the text attack.

1) Sensitivity Analysis of Candidate Substitutions at Different Scales: The choice of the candidate substitution set is a central component of our MOSA-S2 approach. To rigorously evaluate its impact, we conducted experiments using four distinct lexicons of varying sizes.

1) *Prepositions:* A minimal set of 38 English prepositions, establishing a baseline comparable to prior work [6].
2) *SW-10:* A compact set of ten high-frequency stopwords identified through frequency analysis of substitution words in Section V-D.
3) *SW-NLTK:* The standard stopwords list from the NLTK toolkit, serving as the primary lexicon in our main experiments.
4) *SW-spaCy:* An extensive stopwords list from the spaCy library, containing 326 words and approximately doubling the size of the NLTK list.

TABLE VIII

COMPARISON OF SUBSTITUTION STRATEGIES ACROSS DATASETS. "PREPOSITIONS," "SW-NLTK," "SW-SPACY," AND "SW-10" DENOTE CANDIDATE REPLACEMENT SETS BASED ON A SMALL PREPOSITION DICTIONARY, STOPWORD LISTS FROM NLTK AND SPACY, AND TEN HIGH-FREQUENCY STOPWORDS, RESPECTIVELY. "GRAM-CONS" REPRESENTS THE WORD REPLACEMENT SCHEME UNDER GRAMMATICAL CONSTRAINTS

Dataset	Substitution	MC	MR	PPL	Query
MR	Original	99.67%	-	72.70	-
	Prepositions	98.27%	95.88%	114.28	1586
	SW-NLTK	99.93%	96.18%	145.6	5589
	SW-spaCy	99.96%	95.66%	259.18	10798
	SW-10	98.73%	95.85%	100.78	458
	Gram-Cons	92.49%	89.50%	137.15	163
AG News	Original	98.91%	-	62.28	-
	Prepositions	98.29%	96.78%	162.46	3821
	SW-NLTK	99.56%	96.81%	188.89	13893
	SW-spaCy	99.95%	99.95%	308.92	23930
	SW-10	97.05%	96.43%	123.61	1035
	Gram-Cons	97.36%	92.07%	124.13	278
MRPC	Original	94.86%	-	25.86	-
	Prepositions	96.92%	95.67%	87.55	3928
	SW-NLTK	98.01%	95.87%	106.67	12553
	SW-spaCy	98.06%	95.81%	165.84	23875
	SW-10	95.84%	95.74%	81.80	960
	Gram-Cons	87.32%	94.23%	86.03	318

The comparative results, summarized in Table VIII, lead to two principal findings. **First, an expanded candidate lexicon enhances the performance of the rubbish text attack.** As the lexicon size increases from *SW-10* to *SW-spaCy*, the perplexity (PPL) of the generated rubbish examples rises markedly across all datasets. For instance, on the MR dataset, PPL increases from 100.78 to 259.18. Since a higher PPL indicates greater textual confusion and lower fluency, this trend confirms that a larger candidate pool facilitates the creation of more nonsensical and human-unintelligible samples. Crucially, this enhancement in nonsensicality is achieved while the model's confidence (MC) in the original prediction is maintained at a high level, even showing a slight improvement (e.g., MC rises from 98.73% to 99.96%). This verifies that a diverse lexicon is instrumental in generating higher quality rubbish samples that effectively deceive human perception without altering the model's decision.

Second, the results underscore two key strengths of our algorithm: its robustness to the candidate lexicon scale and its fundamental efficiency. As demonstrated in Table VIII, across lexicon sizes ranging from the minimal *SW-10* set to the extensive *SW-spaCy* set, the MR remains consistently high, whereas the MC shows only marginal improvement. For instance, on the MRPC dataset, MC increases merely from 95.84% to 98.06%, with MR staying stable at approximately 96%. This indicates that the algorithm's effectiveness does not depend on a large candidate set. Moreover, the ability of MOSA-S2 to identify high-quality solutions with very few queries when using the compact *SW-10* set (i.e., only 960) highlights its fundamental efficiency. The substantial increase in query count (i.e., 23 875 for *SW-spaCy*) observed with larger lexicons is thus directly attributable to the expanded search space rather than any inherent limitation of the algorithm, further affirming its suitability for resource-constrained scenarios.

2) Rubbish Attacks With Grammatical Constraints: To enhance linguistic plausibility and address perturbation budget concerns, we propose a grammatically constrained replacement strategy. This approach shifts the objective from generating purely nonsensical text to producing *semantically divergent yet grammatically coherent* sentences, thereby evaluating model robustness under more realistic and stealthy attack scenarios.

Our *grammatical constraints (Gram-Cons)* replacement scheme operates as follows.

1) For **nouns and verbs**, we employ a semantic-distance-based replacement strategy that selects the top-k candidates with the lowest cosine similarity to the original word while maintaining identical part-of-speech tags. The candidate pool is primarily drawn from WordNet and supplemented by spaCy when necessary. Each replacement word then undergoes morphological processing to maintain grammatical consistency within the sentence.

TABLE IX

RUBBISH EXAMPLES CRAFTED BY THE SUBSTITUTION WITH GRAMMATICAL CONSTRAINTS. BLUE WORDS INDICATE REPLACEMENTS

Example 1. Dataset: MR Model: BERT Label: Positive (100%)
the casting of von sydow . . . is itself intacto's luckiest stroke.
Gram-Cons: Positive (96%)
the twirling of von sydow . . . is itself intacto's unlucky vendue.
Example 2. Dataset: MR Model: BERT Label: Positive (100%)
yeah, these flicks are just that damn good. isn't it great?
Gram-Cons: Positive (100%)
yeah, these communions are just that bless evilness. isn't it whippy?
Example 3. Dataset: SNLI Model: BERT Label: Entailment (99%)
Premise: Woman in white in foreground and a man slightly behind walking with a sign for John's Pizza and Gyro in the background. Hypothesis: The woman is waiting for a friend.
Gram-Cons: Entailment (100%)
Hypothesis: The retrofit is satirizing for a quantization.

2) For **adjectives and adverbs**, we prioritize replacement with their antonyms from WordNet, constrained by the same top-k selection. If no antonym exists, the strategy defaults to the semantic-distance-based strategy used for nouns and verbs.

Notably, while our method does not guarantee perfect grammaticality, it enforces strict part-of-speech and morphological constraints, significantly improving the coherence and plausibility of generated samples compared to unconstrained stopword substitution. In our experiments, k is set to 10 to align with the scale of the *SW-10* lexicon.

As shown in Table VIII, this grammatically constrained strategy achieves a superior tradeoff between attack efficiency and textual plausibility. By restricting the candidate set to a top-k list, query counts drop dramatically, for example, from 13 893 for *SW-NLTK* to just 278 on the AG News dataset. Although MR decreases predictably and MC is slightly lower than the best stopword-based results, it remains high (e.g., 97.36% on AG News). More importantly, as evidenced by the examples in Table IX, the generated samples exhibit significantly improved readability and grammatical coherence. For instance, in Example 1, replacing "luckiest" with "unlucky" maintains stronger grammatical coherence while introducing explicit semantic opposition. This highlights another crucial advantage: the constrained replacements create a greater semantic deviation from the original text while maintaining MC. Consequently, humans are easily misled by these coherent yet semantically inverted cues to predict different labels, whereas the models persistently maintain their original predictions with high confidence. This clear divergence highlights a fundamental gap between human and model interpretation. Our grammatically constrained variant broadens the scope of rubbish example generation and is especially suited to scenarios where textual plausibility is critical.

TABLE X

ATTACK RESULTS ON LLMS USING MOSA-S2. MC_ORI REPRESENTS THE MODEL'S AVERAGE CONFIDENCE ON THE ORIGINAL SAMPLES AND ΔPPL DENOTES THE INCREMENT IN PERPLEXITY OF THE ATTACKED TEXT RELATIVE TO THE ORIGINAL TEXT

Model	Dataset	MC_Ori	MC	MR	ΔPPL
Vicuna-7B	AG News	86.19%	89.58%	98.68%	128.39
	SNLI	77.15%	81.48%	95.95%	4.15
	MRPC	78.88%	80.43%	96.18%	82.36
Llama-2-Chat-7B	AG News	95.44%	98.30%	98.16%	130.44
	SNLI	99.62%	99.69%	98.48%	6.12
	MRPC	92.60%	96.39%	96.24%	70.64
GPT-3.5	AG News	82.20%	80.62%	80.51%	110.58
GPT-4o	AG News	86.59%	79.65%	67.89%	105.26

VI. ATTACK PERFORMANCE ON LLMS

The deployment of LLMs in safety-critical domains has made their adversarial robustness a paramount concern [20], [21], [22]. While significant research has focused on the oversensitivity of LLMs [19], [20], [48], namely their vulnerability to adversarial examples where minor perturbations like character-level typos cause incorrect predictions, the opposite pathology of undersensitivity remains unexplored in the LLM domain. To fill this research gap, we apply the proposed MOSA-S2 algorithm to LLMs, investigating whether they suffer from the same overconfidence on semantically meaningless text as DNNs do.

The experiments involve open-source LLMs, including Llama-2-Chat-7B and Vicuna-7B, as well as closed-source LLMs, GPT-3.5 (gpt-3.5-turbo-1106), and GPT-4o (gpt-4o-2024-08-06). For Llama-2 and Vicuna, we conduct experiments on 200 randomly selected instances from each of the AG News, SNLI, and MRPC datasets. For GPT-3.5 and GPT-4o, we evaluate on 100 instances from the AG News dataset. The models are prompted to classify the original and attacked texts, and we access GPT-3.5 and GPT-4o through the API offered by OpenAI.

The attack results are summarized in Table X. Our findings are as follows. First, undersensitivity proves to be a prevalent vulnerability, particularly for open-source LLMs. Both Vicuna-7B and Llama-2-Chat-7B exhibit extreme insensitivity, retaining high prediction confidence even when more than 95% of the input words have been substituted. In some cases, MC is observed to increase after the attack, as exemplified by Llama-2's rise from 95.44% to 98.30% on the AG News dataset. This indicates that these models do not rely on a coherent understanding of textual semantics for their predictions but are likely exploiting superficial statistical patterns that remain intact even after heavy perturbation. Second, while the undersensitivity pathology remains observable in closed-source LLMs, they nonetheless exhibit markedly greater robustness than the open-source LLMs, with GPT-4o being particularly robust. For instance, GPT-4o achieves a substantially lower word modification rate (i.e., 67.89%) and exhibits a noticeable drop in MC (i.e., from 86.59% to 79.65%). This suggests that the advanced training paradigms and safety mechanisms employed in closed-source models imbue them with a superior, though

TABLE XI
EVALUATION OF INTERNAL SIMULATION STEPS K ON THREE DATASETS

Dataset	Metric	K = 1	5	10	15	20
MR	MC	98.99%	98.91%	99.96%	99.94%	99.90%
	MR	91.86%	93.63%	95.01%	95.15%	95.12%
AG News	MC	98.70%	99.40%	99.21%	99.55%	99.50%
	MR	88.48%	95.89%	96.81%	97.24%	97.21%
IMDB	MC	98.98%	99.02%	99.91%	99.96%	99.92%
	MR	82.35%	89.91%	91.98%	91.98%	91.87%

TABLE XII
ATTACK RESULTS BY SIMULTANEOUSLY TUNING THE INITIAL TEMPERATURE T_0 AND THE ATTENUATION FACTOR α

α	Metric	T_0 = 50	400	800	1000	1500
0.80	MC	95.52%	97.86%	98.92%	99.40%	99.28%
	MR	94.86%	95.39%	97.81%	96.73%	96.70%
0.85	MC	96.87%	98.53%	99.44%	99.52%	99.32%
	MR	95.88%	96.02%	97.28%	97.02%	96.70%
0.90	MC	96.64%	98.27%	99.20%	99.55%	99.55%
	MR	95.68%	96.23%	97.26%	97.24%	97.15%
0.95	MC	96.65%	98.66%	99.52%	99.27%	99.58%
	MR	95.97%	95.96%	96.30%	96.67%	96.29%

TABLE XIII
ATTACK RESULTS WITH DIFFERENT SAMPLING RADIUS δ

δ	1	2	3	4	5
MC	99.15%	99.55%	99.48%	99.48%	98.95%
MR	96.95%	97.24%	97.24%	96.92%	95.25%

still imperfect, capacity to recognize semantically implausible inputs.

Overall, the experimental results suggest that undersensitivity is also observable in LLMs. The fact that models produce high-confidence predictions for nonsensical text reveals a gap between their statistical proficiency and genuine language comprehension. This vulnerability poses potential security risks: malicious actors could exploit this insensitivity to generate nonsensical or misleading content that LLMs process as valid, leading to critical failures in practical applications. Consequently, we argue that testing for undersensitivity is a necessary component of a comprehensive robustness audit for LLMs, complementing existing research on jailbreaking [21], [22] and text adversarial attack [19], [20].

VII. PARAMETER TUNING

This section details the parameter tuning results for the core parameters, including internal simulation steps K, initial temperature T_0, annealing factor α, sampling radius δ, and the parameter β in the Metropolis principle. All tuning experiments use MC and MR as the primary evaluation metrics. For each dataset, we randomly select 500 instances. During parameter tuning, we find that the same parameter setting can generalize well to various datasets and victim models.

A. Internal Simulation Steps

The internal simulation steps K must be sufficient for the algorithm to approach equilibrium at each temperature. Since the number of replaceable tokens varies with text length, the chosen value of K must be universally adequate across inputs. An overly large K incurs redundant computation for short texts, whereas an insufficient K fails to fully explore the search space for long texts. To identify a robust setting, we sweep K from 1 to 20 and evaluate attack effectiveness on a BERT victim model for short (MR), medium-long (AG News), and long (IMDB) datasets. As shown in Table XI, the MR increases for $K < 15$ and then plateaus, while the MC remains consistently high. Consequently, we set $K = 15$ as the default, striking an optimal balance between attack effectiveness and computational efficiency.

B. Initial Temperature and Attenuation Factor

The highest temperature T_0 and the lowest temperature $T_{\min}$ represent the starting point and the termination condition of the annealing process, respectively. To ensure sufficient iterations while avoiding unnecessary computational overhead, $T_{\min}$ is not fixed but adaptively determined based on the input text length n, defined as $T_{\min} = \alpha^n \cdot T_0$. This design ensures that the number of annealing iterations scales appropriately with the text length, while also reducing the burden of manual temperature tuning. The annealing factor α and the initial temperature T_0 are tightly coupled and jointly determine the tradeoff between exploration and convergence. Specifically, α controls the cooling rate while T_0 sets the initial exploration strength. Therefore, we determine the optimal values for this coupled pair (T_0, α) via a grid search on the AG News dataset using a BERT victim model. The search space spans $T_0 \in [50, 1500]$ and $\alpha \in \{0.80, 0.85, 0.90, 0.95\}$. The results, summarized in Table XII, indicate that attack performance stabilizes once T_0 exceeds approximately 800. Within this stable region, the combination of $T_0 = 1000$ and $\alpha = 0.90$ yields the best balance, achieving a high MR while maintaining superior MC. This combination is therefore selected as the default setting.

C. Sampling Radius

The sampling radius parameter δ regulates the interval for importance-based word sampling at each temperature. At the initial stage ($t = 0$), it enables modification of the least important words, and the radius gradually expands with increasing t. We vary δ from 1 to 5 and evaluate the attack performance against the BERT model on the AG News dataset. The results are presented in Table XIII. As shown, when $\delta \leq 2$, the MR increases with the radius. In contrast, a larger δ leads to a slight decline in MC, indicating that an excessively large search neighborhood introduces noise and hinders effective convergence. To balance these effects and ensure strong cross-dataset performance, we set $\delta = 2$.

TABLE XIV
ATTACK RESULTS WITH DIFFERENT β IN THE METROPOLIS PRINCIPLE

β	80	90	100	110	120
MC	98.89%	99.28%	99.55%	99.55%	99.37%
MR	96.99%	96.95%	97.24%	97.02%	97.30%

D. Metropolis Principle

The parameter β plays a central role in balancing the scale between the nondominated rank differences and the temperature in the Metropolis principle [see (10)]. This scaling is crucial for generating meaningful acceptance probabilities p throughout the annealing process, thereby balancing global exploration and local exploitation. Considering the variation ranges of nondominated rank differences and temperature, we tune β within the range from 80 to 120 with a step size of 10. The evaluation is also conducted on the AG News dataset, with BERT as the victim model. Experimental results listed in Table XIV indicate that setting $\beta = 100$ provides a robust choice.

VIII. CONCLUSION

In this work, we have proposed a novel rubbish text attack algorithm named MOSA-S2. The MOSA-S2 replaces semantically significant words with nonsensical stopwords to craft the text rubbish examples. We also design a multiobjective simulated annealing method to adaptively optimize the word substitution priority, which has a controlled probability of skipping local optima and exploiting Pareto dominance to balance multiple objectives. Extensive experimental results illustrate that the MOSA-S2 outperforms the baselines in the overall metrics. Additionally, better-quality rubbish examples show better properties in model retraining and transfer attack experiments. Our research exposes the excessive insensitivity of text classifiers to semantically meaningful variations: they can maintain more confident original predictions on meaningless word sequences. In the future, our work may be extended to test the robustness of pretrained language models and interpretation methods. Moreover, mitigating this undersensitivity to ensure robust decisions and exploring the tradeoff between excessive sensitivity and undersensitivity are promising directions for future work.

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